Standard Operating Procedure for Resolving Major Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Ericsson PE Role in Basavanagudi POP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_3993

Description

The failure domain of ASIC memory parity errors, line card crashes, and power supply redundancy loss on Ericsson PE role in Basavanagudi POP is attributed to the inherent complexities of the Advanced Switching Architecture (ASA) and the Line Card Module (LCM) in the Ericsson 5620 SRP-21c. The ASA, being a complex multi-layered protocol suite, is susceptible to memory parity errors due to bit flips or other forms of memory corruption. Furthermore, the LCM, being a critical component of the PE role, is prone to crashes if subjected to excessive thermal stress, electrical surges, or other environmental stresses. Additionally, the power supply redundancy loss can occur due to a faulty Power Entry Module (PEM) or a misconfigured power redundancy setting. In such scenarios, the PE role may fail to forward traffic, resulting in a major alarm (ALARM_3993) being raised.

Pre-requisite Checks

- Verify the site location and Node ID (BAS-PE-10) to ensure the correct equipment is being worked upon.

- Check the power redundancy status using the command 'show redundancy power' to ensure that the power supply redundancy is operational.

- Verify the thermal status of the Line Card Module using the command 'show temperature' to ensure that the thermal conditions are within the acceptable range.

- Dispatch remote hands to Basavanagudi POP to perform the necessary physical checks and repairs.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to bit flips or other forms of memory corruption	High
Line Card Module crash due to excessive thermal stress or electrical surges	Medium
Power supply redundancy loss due to faulty Power Entry Module or misconfigured power redundancy setting	Low

Diagnostic Commands & Expected Outputs

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Command: show system memory, Expected Output: Memory parity error detected, Description: This command will verify if there is a memory parity error in the system, which can be a root cause of the failure.
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```
Command: show temperature, Expected Output: Temperature above threshold, Description: This command will verify if the thermal conditions of the Line Card Module are within the acceptable range, which can be a root cause of the failure.

Command: show redundancy power, Expected Output: Power supply redundancy lost, Description: This command will verify if the power supply redundancy is operational, which can be a root cause of the failure.
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Resolution Steps

Step 1: Step: Power cycle the Line Card Module, Command: shutdown linecard , Description: Power cycling the Line Card Module may resolve the issue if it is caused by a software or hardware malfunction.

Step 2: Step: Replace the faulty Power Entry Module, Command: replace power-entry-module , Description: Replacing the faulty Power Entry Module may resolve the issue if it is caused by a faulty Power Entry Module.

Step 3: Step: Reset the system to default configuration, Command: reset system default, Description: Resetting the system to default configuration may resolve the issue if it is caused by a misconfigured system.

Post-Check Verification

```
Command: show system memory, Expected Output: No memory parity error detected, Description: This command will verify if the memory parity error is resolved after the resolution steps.

Command: show temperature, Expected Output: Temperature within threshold, Description: This command will verify if the thermal conditions of the Line Card Module are within the acceptable range after the resolution steps.

Command: show redundancy power, Expected Output: Power supply redundancy operational, Description: This command will verify if the power supply redundancy is operational after the resolution steps.
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Roll-back Procedures

- Step: Power cycle the system, Command: shutdown system, Description: Power cycling the system will ensure that the system is reverted to its original state before the resolution steps.

- Step: Restore the system configuration, Command: configure system default, Description: Restoring the system configuration will ensure that the system is reverted to its original state before the resolution steps.